

CareBot Final ESP32 Model

Three medical-kit bins and two beam grippers

Hardware and route matched to the final firmware

Purpose

CareBot carries ten preloaded medical kits and two containment beams. Three gravity bins hold groups of 2, 6, and 2 kits. A servo opens each flap once and leaves it open. Two more servo grippers hold one beam across the front of the robot and one along its right side. At the final stop, both grippers open and the robot remains stopped.

The robot starts in the top-right area facing left. It uses a front ultrasonic sensor to stop near walls and two rear downward-facing IR sensors to confirm the black crossing near the middle delivery area. Turns and short corrections use calibrated motor timing.

Final hardware

Component	Quantity	Job
DOIT ESP32 DEVKIT V1	1	Main controller
L298N dual motor-driver module	1	Controls both drive motors
Dual-shaft 12 V, 500 RPM DC geared motor	2	Left and right drive
Front ultrasonic sensor	1	Wall-distance measurement
Rear digital IR reflectance sensor	2	Paired black-crossing detection
PCA9685 16-channel servo driver	1	Controls all five servos over I ² C
Positional servo	5	Three bin flaps and two beam grippers
Start pushbutton	1	Starts one mission after reset
Drive wheels and caster	2 + 1	Supports differential drive
Main power switch	1	Disconnects the battery

The motors are rated for 12 V. The L298N module must handle each motor's measured startup and stall current. Its bridge also drops some voltage, so the motors receive less than the battery voltage while running. The servo supply must be regulated and sized for five servos. Servos must not draw power through the ESP32 board. All supplies and modules share ground.

ESP32 pin map

GPIO	Connection
25, 26	L298N IN1, IN2 for the left motor
27, 14	L298N IN3, IN4 for the right motor
33, 17	L298N ENA, ENB motor PWM
23, 34	Ultrasonic TRIG, ECHO
35, 16	Left and right rear IR digital outputs
21, 22	PCA9685 SDA, SCL
32	Start button to ground, using the internal pull-up

GPIO34 and GPIO35 are input-only pins, which suits the ultrasonic ECHO and left IR inputs. A 5 V ultrasonic ECHO signal needs a divider before GPIO34. The planned divider is 10 k Ω from ECHO to GPIO34 and 15 k Ω from GPIO34 to ground. Both IR outputs must remain within the ESP32's 3.3 V input limit. Remove the L298N ENA and ENB jumpers so GPIO33 and GPIO17 can control speed.

Connect PCA9685 VCC to ESP32 3.3 V, OE to ground, and leave its address at the default 0x40. Connect the regulated servo supply to V+ and join all grounds. Channels 0–4 drive the two-kit bin, six-kit bin, two-kit bin, right beam gripper, and front beam gripper respectively. Never power the servos from ESP32 3.3 V or PCA9685 VCC.

Mission sequence

1. Start at the top right facing left. Drive toward the left wall and stop at the calibrated ultrasonic clearance.
2. Turn left to face down. Open the first flap and drop two kits near the top-left destination.
3. Drive down the left side. Confirm the first valid transverse black crossing with both rear IR sensors, apply the calibrated outlet correction, and open the middle flap to drop six kits.
4. Continue toward the bottom wall. Stop at the calibrated clearance and open the third flap to drop the last two kits.
5. Turn left to face right. Apply the optional lane correction, then drive toward the right wall beside the beam area.
6. Stop at the single calibrated beam-drop position. Open both beam grippers, wait for the beams to fall, and remain stopped.

The code releases a complete preloaded group; it does not count individual kits. The first black separator indicates entry to the middle area, so the rear sensor position and bin outlet position determine the correction after detection.

Firmware behavior

The only sketch is `output/CareBot/CareBotESP32/CareBotESP32.ino`. Select **DOIT ESP32 DEVKIT V1** in Arduino IDE and use the Espressif Arduino-ESP32 3.x board package. No separate servo library is required. The detailed board-label map is in `output/CareBot/WIRING.md`.

On boot, the program detects the PCA9685, configures it for 50 Hz, commands all five servo channels closed, and runs a startup self-check. It verifies unique GPIO assignments, motor PWM setup, PCA9685 response, servo commands, one valid ultrasonic echo, both IR input states, and the START input state. A missing driver, failed I²C command, or missing ultrasonic echo blocks the mission. The start button must first be released and then held for at least 40 ms. One press starts one mission; reset is required before another run. Sending x through Serial Monitor at 115200 baud stops the motors while USB remains connected. The main switch provides the physical power stop.

Serial messages contain a millisecond timestamp and an INFO, WARN, or ERROR severity. Every reported error disables both L298N enable outputs. The program warns that it cannot verify motor rotation, driver current, or servo position because those devices have no feedback connections.

The program stops on missing ultrasonic echoes, wall-target overshoot, missing markers, travel timeout, invalid settings, invalid servo commands, or a motor PWM failure. It requires three stationary wall readings within tolerance before accepting a wall stop. It does not have sensors to verify an empty bin or an upright beam.

Values that must be measured

Before a field run, measure and update motor direction, loaded forward and reverse speed, left-turn duration, servo closed and open angles, wall clearances, the middle outlet correction, beam-lane shift, and the final beam-drop wall distance. The supplied values are examples. A 500 RPM motor label does not determine travel speed without the wheel diameter and loaded motor speed.

The front ultrasonic sensor must see past or below the carried front beam. Both rear IR sensors must see black together to confirm a crossing, but they do not continuously steer along a line. Reverse movement has no rear obstacle sensor, and turns have no side-clearance sensor, so the loaded route must be checked physically.

Verification status

The current L298N and PCA9685 source compiled for `esp32:esp32:esp32doit-devkit-v1` with Arduino-ESP32 3.3.5 and all warnings enabled. It also passes a warning-clean C++17 compile against the host regression harness. The tests cover PCA9685 detection and channel writes, motor direction, startup sensor failure, wall overshoot, start debounce, marker handling, the full 2–6–2 route, open-only flaps, and stationary release of both beams. No physical robot run has been completed, so mechanical and navigation performance remain unverified.

Second robot status

The second robot has not been built. Its retained future plan covers sample collection and delivery to the laboratory. That plan is stored separately in `docs/SECOND-ROBOT-FUTURE.md` so it cannot be confused with this final CareBot design.